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Lecture 23-24

Intro to Statistical Learning

→ study here.

- $Y = \beta_0 + \beta_1 X + \varepsilon$
  ↳ marks in stats. inference $H_0 : \beta_1 = 0$, $H_1 : \beta_1 \neq 0$
  in predictions → $\hat{Y}_{new} = \hat{\beta}_0 + \hat{\beta}_1 X_{new}$
  ⇒ most interpretable model is not always the most accurate one.

Inference	Learning
θ	$f(\cdot)$
$\hat{\theta}(D)$	$\hat{f}(\cdot; D)$
distribu^n of $\hat{\theta}$	distri. of fitted predictions
MSE	predictive error
MLE	empirical risk mini.

- in stat. learning, we estimate a func^n $f : X \to Y$
- 2 RV → X: input, Y: response
  $Y | X = x$
  (this depends on how we measure the cost of being wrong.
  Loss func^n → $L(Y, a)$
- for a fixed input x, the conditional loss is — $r(a|x) = E[L(Y,a) | X = x]$
  a, that minimises this expected loss.

$$f^*(x) = \arg\min_a E[L(Y,a) | X = x]$$

- $L(Y,a) = (Y-a)^2$

$$f^*(x) = \arg\min_a E[(Y-a)^2 | X = x]$$

$$\mu(x) = E[Y | X = x]$$

$$Y - a = (Y - \mu(x)) + (\mu(x) - a) \to \text{sq.}$$

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$$E[(Y-a)^2 \mid X=x] = Var(Y \mid X=x) + (\mu(x)-a)^2$$

$$f^*(x) = E[Y \mid X=x]$$

under squared loss, the optimal prediction is the conditional mean.

Eg

Y	0	5	10
P	0.2	0.5	0.3

$$E[Y \mid X=x] = 0 + 2.5 + 3 = 5.5$$

$$f^*(x) = 5.5$$

- now, what if $g(a) = E[|Y-a| \mid X=x]$ → absolute loss

$$P(Y \le a \mid X=x) = 0.5$$

$$f^*(x) = \text{Median}(Y \mid X=x)$$

- The median performs better under abs. loss as the outlier does not pull it toward 1 side strongly. Sq. loss penalizes large errors quadratically. Abs. loss penalizes errors linearly. ∴, abs. loss is preferable when outliers should have less influence.

- 0-1 loss → Bayes classifier

$Y \in \{1, 2, \ldots, K\}$ $\quad L(Y,a) = \begin{cases} 0 & Y = a \\ 1 & Y \neq a \end{cases}$

$$f^*(x) = \arg\max_{k \in \{1,\ldots,K\}} P(Y=k \mid X=x)$$

↳ Bayes classifier

Eg. $P(Y=1 \mid x) = 0.15$

$P(Y=2 \mid x) = 0.65$ ✓

$P(Y=3 \mid x) = 0.2$

$f^*(x) = 2$

error prob. $= 1 - 0.65 = 0.35$

- Asym. loss → classifica" threshold
→ why we should not classify obs. using the threshold of 0.5.

$$\boxed{p(x) > \frac{C_{FP}}{C_{FP} + C_{FN}}}$$

0.5 is not a universal law of classifica". It happens when you assign equal costs to FP & FN.

- sq. loss	$(Y-a)^2$	conditional mean
abs. loss	$\lvert Y-a \rvert$	conditional median
0-1 loss	$1(Y \neq a)$	most probab. class
asym. binary classifica" loss	unequal FP FN costs	cost dependent threshold

- Statistical Risk → risk of a predictor f is its expected loss on a new obs.

$$\boxed{R(f) = E_{(X,Y)}\left[L(Y, f(X))\right]}$$

Now, the expectation is taken over both X & Y. In conditional risk we had fixed $X = x$. so by law of total expectation,

$$R(f) = E_X\left[E\left[L(Y, f(X)) \mid X\right]\right]$$

the risk measures the average predic" cost over the population

- Bayes' Risk → let f^* be the optimal predic" func" over all measurable predictors for which the expected loss is defined.

$$\boxed{R^* = R(f^*) = \inf_f R(f)}$$

↳ Bayes' Risk

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- Irreducible error under sq. loss.

$$Y = f^*(X) + \varepsilon \qquad f^*(X) = E[Y|X]$$

$\because$, $E[\varepsilon|X] = 0$

$\text{Var}[\varepsilon|X] = \sigma^2$

we want $E\left[\left(Y - \hat{f}(X)\right)^2 \mid X = x\right]$

$$\boxed{E\left[\left(Y - \hat{f}(X)\right)^2 \mid X = x\right] = \underbrace{\left(\hat{f}(x) - f^*(x)\right)^2}_{\text{reducible error}} + \underbrace{\sigma^2}_{\text{irreducible error}}}$$

We can reduce reducible error by using a better model, more useful predictors, improved estimation & more training data.

Eg. $Y = 3X + 2 + \varepsilon$, $E[\varepsilon|X] = 0$, $\text{Var}(\varepsilon|X) = 4$

$\hat{f}(x) = 2.5x + 1$

$x = 2$

$f^*(2) = 3(2) + 2 = 8$

$\hat{f}(2) = 2.5(2) + 1 = 6$

$\therefore$, conditional expected sq. error $= (6-8)^2 + 4 = 8$

- In theory, $f^* = \arg\min R(f)$. But we don't know the true probability distribu^n of (X, Y). Therefore, we restrict our attent^n to func^n class. We say func^n class F. Then estimate $\hat{f} \in F$, using the training data.
- The excess risk of a fitted predictor is $R(\hat{f}) - R(f^*)$

$$\boxed{R(\hat{f}) - R(f^*) = \underbrace{\left[\inf_{f} R(f) - R(f^*)\right]}_{\text{approxima}^n \text{ error}} + \underbrace{\left[R(\hat{f}) - \inf_{f} R(f)\right]}_{\text{estima}^n \text{ error}}}$$

approxima^n error: measures the gap btw. Bayes optimal predictor & best available predictor

estima^n error: measures the diff. btw. the risk of the fitted predictor & that of the best predictor

Eg. $R(f^*) = 2$, $\inf R(f) = 5$, $R(\hat{f}) = 7$

approx error = $5 - 2 = 3$
estimat^n error = $7 - 5 = 2$ ∴
total excess risk = $3 + 2 = 5$

- Let's say $F_1 \subseteq F_2$, F_2 has more candidate func^n

$$\inf_{F_2} R(f) \leq \inf_{F_1} R(f)$$

∴, the approxima^n error doesn't increase when func^n class is enlarged. A larger func^n class can make learning from finite data more difficult. Thus for overfit.

- Linear model

$$f(x) = \beta_0 + \sum_{j=1}^{p} \beta_j x_j$$

predictor is linear in its coeff.

- Sparse model $f(x) = \beta_0 + \sum_{j=1}^{p} \beta_j x_j$

with many coeff. = 0,

lasso → $\lambda \sum_j |\beta_j|$ } sparsity

rigde → $\lambda \sum_j \beta_j^2$ } shrinkage

- Generalised additive models

$$f(x) = \beta_0 + g_1(x_1) \dots + g_p(x_p)$$

each feature might be non-linear, but predictor has additive structure.

- decision tree → partition the feature space into reg. & assign a predic^n to each reg.

$$f(x) = \sum_{m=1}^{M} e_m \, 1(x \in R_m)$$

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- RF → combine DT, averaging their predict^n

$$\hat{f}_{RF}(x) = \frac{1}{B} \sum_{b=1}^{B} T_b(x)$$

- SVM → uses decision func^n in a feature space, with margin-based learning

$$f(x) = w^T \phi(x) + b$$

- EDA → helps in inspecting & summarizing at dataset before building predictive models.

Mean → central tendency
Median → robust " "
SD → variability
histogram → distribu^n shape, skewness
boxplot → spread & potential outliers
scatterplot → relationship btw. variables
class-freq. → class imbalance

- Anscombe's quartet → why summary stats can be misleading.